

Task 2: Simulate Banker's Algorithm used for Deadlock Avoidance and find whether the system is in safe state or not.

```
#include <stdio.h>
```

```
int main() {
    int p,r;
    printf("Enter the no. of processes: ");
    scanf("%d",&p);
    printf("Enter the no. of resources: ");
    scanf("%d",&r);
    int all[p][r],max[p][r],nee[p][r],ava[r],tot[r],seq[p],com[p],cou=0;
    for(int i=0;i<p;i++)
    {
        printf("\n\nProcess %d \n",i+1);
        for(int j=0;j<r;j++)
        {
            printf("Maximum value for resource %d: ",j+1);
            scanf("%d",&max[i][j]);
        }

        for(int j=0;j<r;j++)
        {
            printf("Allocated from resource %d: ",j+1);
            scanf("%d",&all[i][j]);
        }
    }
    printf("\n\n");
    for(int j=0;j<r;j++)
    {
        printf("Enter total value of resource %d: ",j+1);
        scanf("%d",&tot[j]);
    }
    for(int j=0;j<r;j++)
    {
        ava[j]=tot[j];
    }
    for(int j=0;j<p;j++)
    {
        com[j]=0;
    }
    for(int i=0;i<p;i++)
    {
        for(int j=0;j<r;j++)
```

```

        {
            ava[j]-=all[i][j];
        }
    }
    for(int i=0;i<p;i++)
    {
        for(int j=0;j<r;j++)
        {
            nee[i][j]=max[i][j]-all[i][j];
        }
    }

    while( cou<p)
    {
        int runcheck=0;
        for(int j=0;j<p;j++)
        {
            if(com[j]==0)
            {
                int k;
                for(k=0;k<r;k++)
                {
                    if(nee[j][k]>ava[k])
                        break;
                }
                if(k==r)
                {
                    for(int i=0;i<r;i++)
                    {
                        ava[i]+=all[j][i];
                    }
                    seq[cou]=j;
                    com[j]=1;
                    cou++;
                    runcheck=1;
                }
            }
        }
        if(runcheck==0)
        {
            printf("\n\nThe system is currently in not safe state.\n");
            return 0;
        }
    }
}

```

```

printf("\n\nThe system is currently in not safe state and \n< ");
for(int j=0;j<p;j++)
{
    printf("P%d ",seq[j]);
}
printf("> is the safe sequence.\n");
return 0;
}

```

Input:

Enter the no. of processes: 4

Enter the no. of resources: 3

Process 1

Maximum value for resource 1: 3

Maximum value for resource 2: 2

Maximum value for resource 3: 2

Allocated from resource 1: 1

Allocated from resource 2: 0

Allocated from resource 3: 0

Process 2

Maximum value for resource 1: 6

Maximum value for resource 2: 1

Maximum value for resource 3: 3

Allocated from resource 1: 5

Allocated from resource 2: 1

Allocated from resource 3: 1

Process 3

Maximum value for resource 1: 3

Maximum value for resource 2: 1

Maximum value for resource 3: 4

Allocated from resource 1: 2

Allocated from resource 2: 1

Allocated from resource 3: 1

Process 4

Maximum value for resource 1: 4

Maximum value for resource 2: 2

Maximum value for resource 3: 2

Allocated from resource 1: 0

Allocated from resource 2: 0

Allocated from resource 3: 2

Enter total value of resource 1: 9

Enter total value of resource 2: 3

Enter total value of resource 3: 6

Output:

The system is currently in not safe state and
< P1 P2 P3 P0 > is the safe sequence.